

1 Model

A cost-minimizing government chooses how to allocate a given stock of debt across three creditor sectors—domestic non-banks, foreigners, and the banking sector. The three sectors differ in *how* placement becomes costly at the margin, and those differences produce a tractable pecking order that we take to the data.

1.1 Setting

A government has a given amount of outstanding debt $D > 0$ and chooses non-negative holdings (d_h, d_f, d_b) satisfying

$$d_h + d_f + d_b = D, \quad (1)$$

where d_h is non-bank holdings, d_f is foreign holdings, and d_b is banking (central bank plus private banks).

Every placement is costly.

- **Non-banks and foreigners** lend at interest rates $r_h(d_h, S_h)$ and $r_f(d_f)$; the government's cash cost of placement is the interest bill $r_h d_h + r_f d_f$.
- **Banking** is not summarised by an interest rate: placing d_b with the banking system carries a separate welfare (quasi-fiscal) cost $\Psi(d_b)$ —inflation, monetary/fiscal trade-offs, financial-stability risk.

Structural size. $S_h > 0$ is the *country-specific* non-bank sector size (capacity).

Planner problem. Total debt $D > 0$ is given. Choose $d_h, d_f, d_b \geq 0$ to minimise total placement cost subject to the budget constraint:

$$\min_{d_h, d_f, d_b} r_h(d_h, S_h) d_h + r_f(d_f) d_f + \Psi(d_b) \quad \text{s.t.} \quad d_h + d_f + d_b = D. \quad (2)$$

1.2 The three sectors: how placement becomes costly

Marginal cost rises with the sector's *degree of saturation*. What differs is *how* saturation is measured.

Non-banks.

$$r_h(d_h, S_h) = \bar{r}_h + \gamma_h \frac{d_h/S_h}{\bar{\omega} - d_h/S_h}. \quad (3)$$

Saturation is the *relative amount* d_h/S_h . The rate stays near $\bar{r}_h$ when the sector is far from its cap $\bar{\omega}S_h$ and diverges as $d_h \rightarrow \bar{\omega}S_h$. Larger S_h means more headroom and later saturation. S_h differs across countries.

Foreigners.

$$r_f(d_f) = \bar{r}_f + \gamma_f \frac{d_f}{\bar{D}_f - d_f}. \quad (4)$$

Saturation is the level of foreign exposure d_f . The rate diverges as $d_f \rightarrow \bar{D}_f$ (a no-default ceiling). $\bar{D}_f$ is *common* across countries.

Banking.

$$\Psi(d_b), \quad \Psi'(d_b) > 0, \quad \Psi''(d_b) > 0. \quad (5)$$

There is no hard ceiling, but placement carries a welfare cost that is increasing and convex.

1.3 Equilibrium: who absorbs the next dollar?

At the optimum, the planner equates the marginal cost of placing one more dollar of debt across all three sectors:

$$MC_h(d_h; S_h) = MC_f(d_f) = MC_b(d_b) = \lambda(D), \quad d_h + d_f + d_b = D. \quad (6)$$

If one sector were cheaper at the margin, the planner would shift toward it.

How the equilibrium plays out as D grows.

- **Low D .** Non-banks and foreigners are far from their limits; MC_h and MC_f are low and they absorb most new debt. Banking is costlier at the margin and absorbs little.
- **Rising D .** Non-banks and foreigners approach their limits; MC_h and MC_f rise sharply and their absorption share shrinks. Banking still has room and starts picking up.
- **High D .** Non-banks and foreigners are near their caps; $MC_h, MC_f \rightarrow \infty$ and they absorb almost nothing at the margin. Banking is the only sector with a finite marginal cost and absorbs nearly all new debt.

1.4 Model predictions

1. **Every sector absorbs more debt as D grows.** Upward-sloping MC_j implies that as D rises, all three margins rise together and every d_j^* rises.
2. **Larger non-bank sectors absorb more debt at any given D .** A bigger S_h means a less saturated non-bank sector at any d_h , a lower MC_h at the margin, and so the planner allocates more to non-banks.

3. As D grows, non-banks and foreigners absorb a smaller share of each extra dollar; their marginal shares go to zero. Under symmetry, the paths of d_h^* and d_f^* are strictly concave in D . Both approach caps, so MC_h and MC_f rise and diverge, and the planner shifts each extra dollar toward banking.
4. **Banking takeover at high D .** With non-banks and foreigners near their caps, banking is the only sector with room, and its share of new issuance rises sharply.

1.5 Mechanics I: three phases of absorption

One country, $S_h = 150$. Baseline calibration $\bar{r}_h = \bar{r}_f = \bar{r}_b = 1$, $\gamma_h = \gamma_f = 0.02$, $\bar{\omega} = 0.30$, $\psi = 0.04$, $\bar{D}_f = 55$. Figure ?? shades the three equilibrium phases, defined by banking's marginal share $s_b \equiv dd_b^*/dD$.

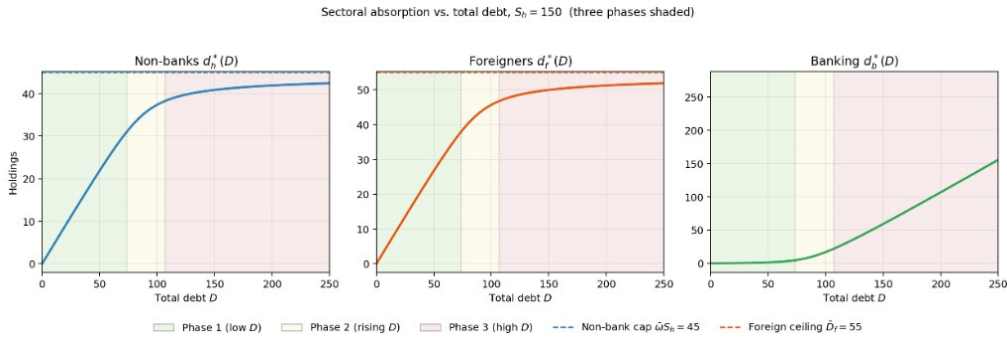


Figure 1 Three phases of debt absorption for a representative country ($S_h = 150$).

- **Phase 1.** Non-banks and foreigners absorb; banking flat.
- **Phase 2.** Non-banks and foreigners saturate; banking starts picking up.
- **Phase 3.** Banking takes over. The transition happens at $D \approx \bar{\omega}S_h + \bar{D}_f = 100$, matching the empirical kink at 100% of GDP documented in Section ??.

1.6 Mechanics II: heterogeneous non-bank sector

Comparative statics in S_h ($S_h \in \{30, 70, 110, 150, 200\}$; $\bar{D}_f$, $\bar{\omega}$, γ_h , γ_f , ψ fixed at the baseline).

Pecking order.

- **Non-banks:** five different saturation levels, one per country, at $\bar{\omega}S_h$.
- **Foreigners:** essentially one curve—all countries approach the *same* ceiling $\bar{D}_f$, differing only in how fast they get there.
- **Banking:** once total debt is large enough, banking takes over for all countries.

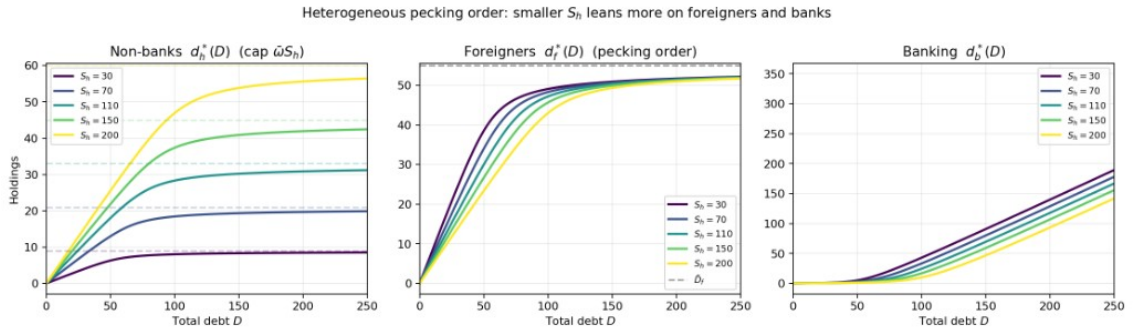


Figure 2 Pecking order across countries with different non-bank sector sizes S_h .